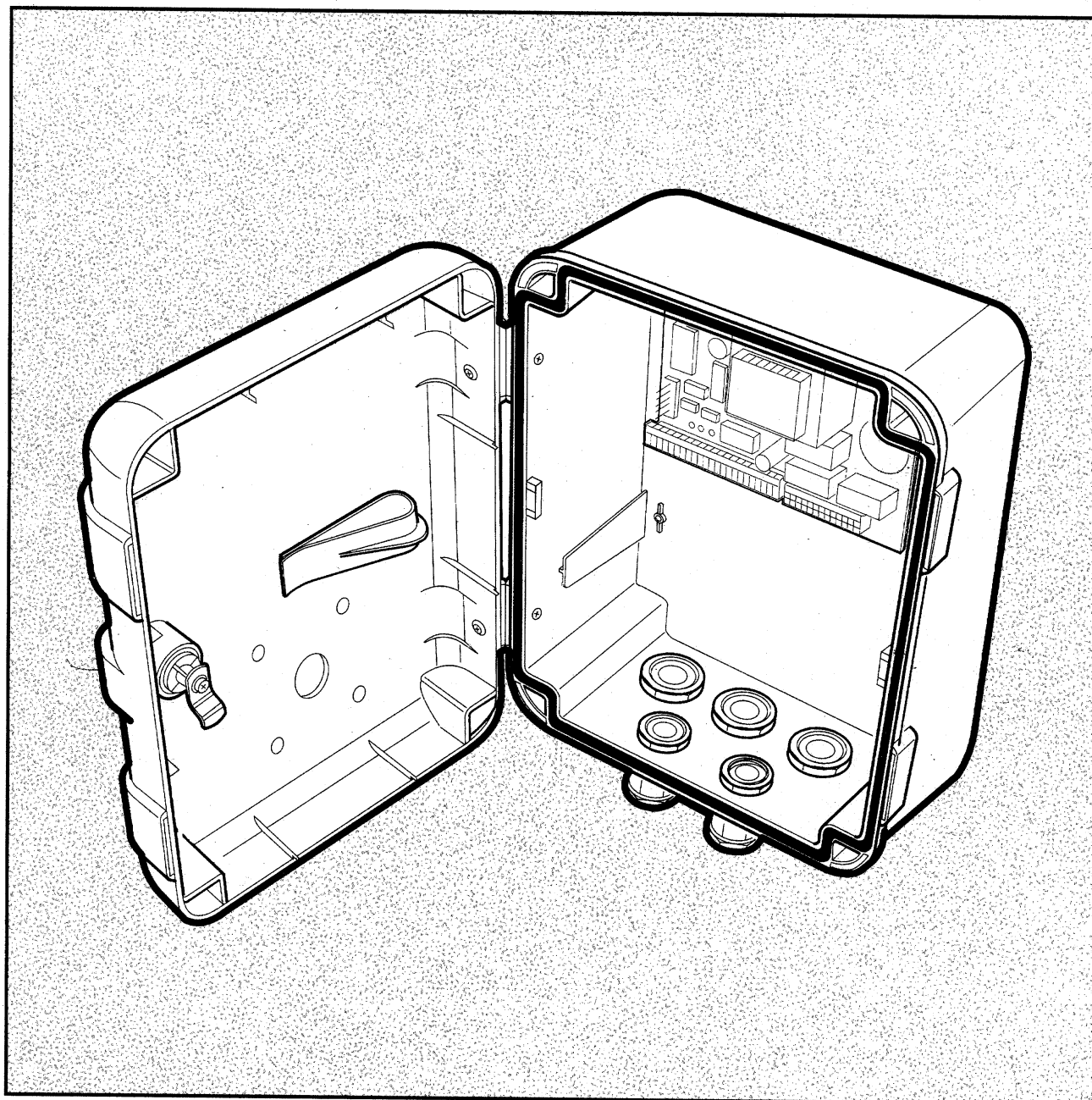


401 MPS



FAAC[®]

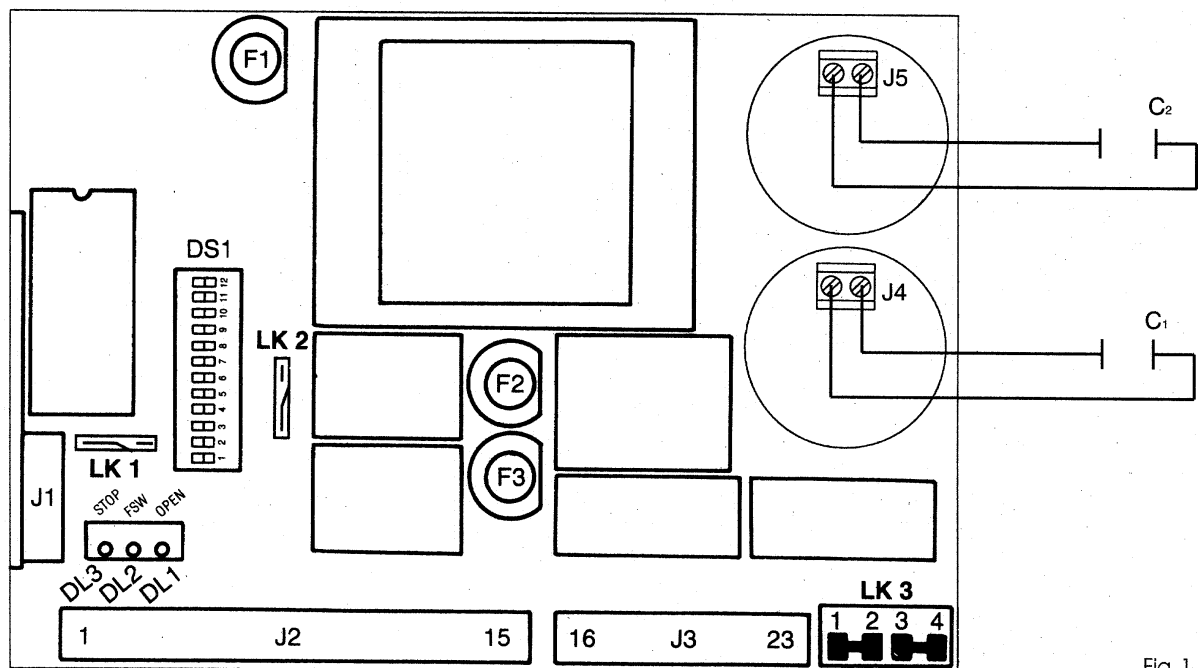


Fig. 1

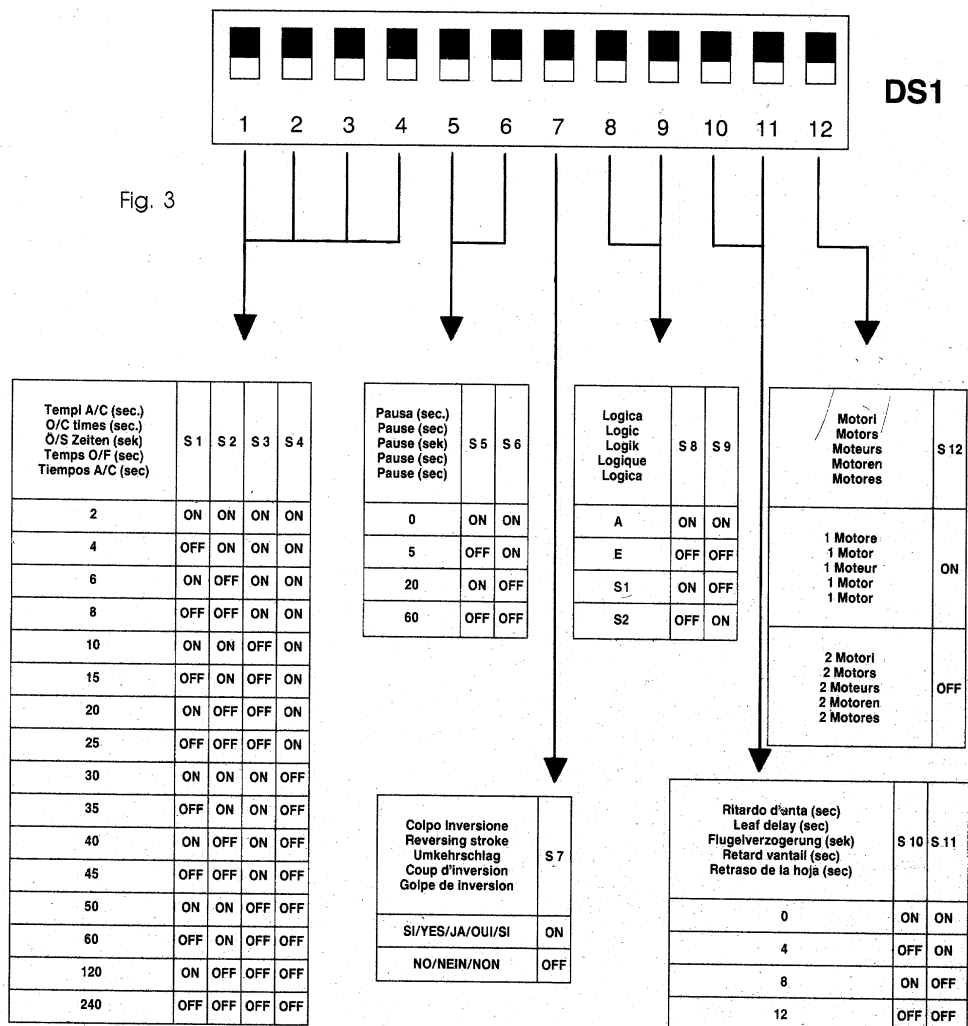
LK1	FUNZIONAMENTO ELETTROSERRATURA LOCK INTERVENTION MODE FUNCTIONEMENT ELECTRO-SERRURE ELEKTROSHLOSSBE TRIEB FUNCIONAMIENTO ELECTROCERRADURA
 OFF	IN APERTURA E CHIUSURA IN OPENING AND CLOSING EN OUVERTURE ET FERMETURE BEI OFFNUNG UND SCHLIESSEN EN APERTURA Y CIERRE
 ON	SOLO IN APERTURA IN OPENING ONLY SEULEMENT EN OUVERTURE NUR BEI OFFNUNG SOLO EN APERTURA

Fig. 2

LK2	FUNZIONAMENTO LAMPADA SPIA WARNING LIGHT OPERATION MODE FONCTIONNEMENT DE LA LAMPE TEMOIN BETRIEBSLOGIKEN DER SIGNALLAMPE FUNCIONAM. DE LA LAMPADA PILOTO
 OFF	ACCESA IN APERTURA/LAMPEGGIANTE IN CHIUSURA LIGHT ON WHEN OPENING/FLASHING WHEN CLOSING ALLUMEE EN OUVERTURE/CLIGNOTANTE EN FERMETURE ANGENACHT BEI OFFNUNG/BEINKEND BEI SCHLIESSEN ENCENDIDA EN APERTURA/INTERMITENTE EN CIERRE
 ON	ACCESA IN APERTURA E CHIUSURA LIGHT ON WHEN OPENING AND CLOSING ALLUMEE EN OUVERTURE ET FERMETURE ANGENACHT BEI OFFNUNG UND SCHLIESSEN ENCENDIDA EN APERTURA Y CIERRE

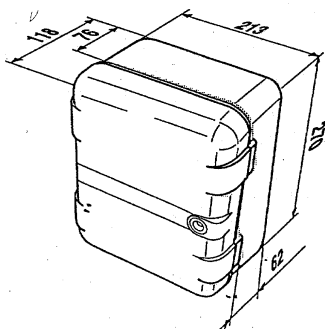
Fig. 4

Fig. 3



DIMENSIONI MAX. D'INGOMBRO
OVERALL DIMENSIONS
DIMENSIONS MAXI. D'ENCOMBREMENT
MAX. PLATZBEDARF
DIMENSIONES EXTERNAS

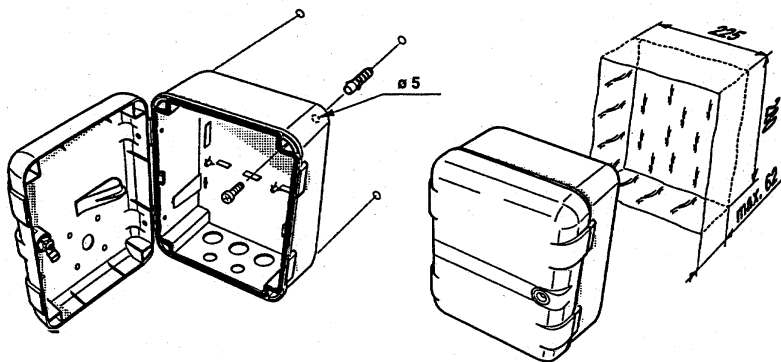
Quote in mm
 Values in mm
 Cotes en mm
 Maße in mm
 Cotas en mm



INSTALLAZIONE • INSTALLATION • INSTALLATION • INSTALLATION • INSTALACIÓN

Fissaggio a parete • Wall mounting • Fixation murale • Aufputzmontage • Montaje de aplique

Fissaggio ad incasso • Inset mounting • Montage encastré • Unterputzmontage • Montaje empotrado



PREDISPOSIZIONI (MATERIALE COMMERCIALE NON DI FORNITURA FAAC) • FITTINGS (COMMERCIAL FITTINGS NOT MANUFACTURED BY FAAC)
PREEQUIPEMENT (LE MATERIEL COMMERCIAL N'EST PAS FOURNI PAR FAAC) • VOREINSTELLUNGEN (HANDELSÜBLICHES MATERIAL NICHT IN FAAC-LIEFERUNG INBEGRIFFEN) • ELEMENTOS NECESARIOS (MATERIAL COMERCIAL NO SUMINISTRADO POR FAAC)

1. Sezionatore rotativo bipolare con dispositivo chiudiporta per montaggio su guida DIN. Riferimenti "sprecher+schuh": sezionatore LA 2-12-1752; prolunga albero modulatore LA 2-G-2853; albero per sezionatore LA 2-G-3195; monopola LFS 2-N-6
2. Morsetto per conduttore di terra per montaggio su guida DIN
3. Interruttore magnetotermico differenziale (portata 10A con soglia d'intervento 0,03A) per montaggio su guida DIN
4. Guida DIN
5. Supporto condensatore
6. Pressacavo PG21
7. Raccordo tubo/cassetta (tubo Ø 25)
8. Pressocavo PG 11

IMPORTANTE: Per mantenere il grado di protezione IP55 del contenitore utilizzare raccordi tubi/cassetta, pressocavi ed eventuali sezionatori rotativi che garantiscono il medesimo grado di protezione.

1. Bipolar rotary switch with door lock for mounting on DIN rail.
Sprecher+Schuh part numbers: switch LA2-12-1752 modular shaft extension LA2-G-2853 switch shaft LA2-G-3195 knob LFS2-N-6
2. Earth (ground) lead terminal for mounting on DIN guide.
3. Differential magnetic-thermal cutout (capacity 10A, trip threshold 0.03A).
4. DIN rail.
5. Capacitor mounting.
6. PG21 cable sleeve.
7. Pipe union (for diam. 25 mm pipes).
8. PG11 cable sleeve.

IMPORTANT: In order to maintain IP55 protection rating, use only pipe unions, cable guides, and rotary switches with the same protection rating.

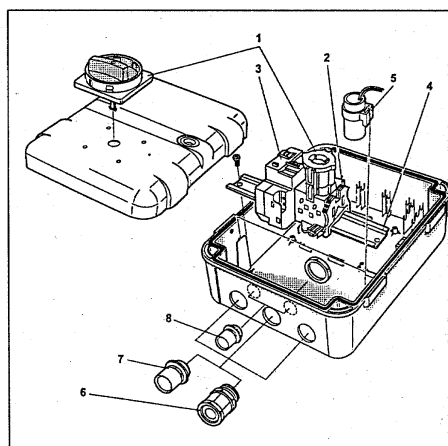
1. Sectionneur rotatif bipolaire avec dispositif de fermeture de porte pour montage sur guide DIN, références "sprecher + schuh": sectionneur LA2 - 12 - 1752 rallonge arbre modulaire LA2 - G - 2853 arbre pour sectionneur LA2 - G - 3195 bouton LFS2 - N - 6
2. Borne pour conducteur de terre pour montage sur guide DIN
3. Disjoncteur magnéto-thermique différentiel (pouvoir de coupure 10A, seuil d'intervention 0,03 A) pour montage sur guide DIN
4. Guide DIN
5. Support condensateur
6. Passe-câble PG21
7. Raccord tube/coffret (tube Ø 25)
8. Passe-câble PG11

IMPORTANT: pour conserver le degré de protection IP55 du coffret, veuillez utiliser des raccords tube/coffret, des passe-câbles et d'éventuels sectionneurs rotatifs qui garantissent le même degré de protection.

1. Zweipoliger Dreh-Trennschalter mit Tor-Schließvorrichtung für Montage auf DIN-Profilschiene. Bezüge "sprecher + schuh": Trennschalter LA 2 - 12 - 1752 Modul-Wellenverlängerung LA 2 - G - 2853 Welle für Trennschalter LA 2 - G - 3195 Handgriff LFS 2 - N - 6
2. Klemme für Erdleiter zur Montage auf DIN-Profilschiene.
3. Fehlerstrom-Leistungsschalter (Aufnahme 10-A mit Schwellwert 0,03 A) für Montage auf DIN-Profilschiene.
4. DIN-Profilschiene.
5. Halterung Kondensator.
6. Kabelhalter PG21.
7. Anschluß Leitung/Gehäuse (Leitung Durchm. 25)
8. Kabelhalter PG11.

WICHTIG: Zur Gewährleistung der Gehäuse-Schutzart IP55 müssen die verwendeten Anschlüsse Leitung/Gehäuse, die Kabelhalter und eventuelle Dreh-Trennschalter die gleiche Schutzart aufweisen.

1. Seccionador rotativo bipolar con dispositivo de cierre de la puerta para montar en guía DIN.
Referencias Sprecher + Schuh: seccionador LA 2 - 12 - 1752 prolongación eje modular LA 2 - G - 2853 eje para seccionador LA 2 - G - 3195 perilla LFS 2 - N - 6
2. Borne para conductor de tierra para montar en guía DIN.
3. Interruptor magnetotérmico diferencial (capacidad 10 A con sensibilidad de intervención de 0,03 A) para



montar en guía DIN.

4. Guía DIN.
5. Soporte para el condensador.
6. Prensacable PG21.
7. Racor tubo/caja (tubo f 25).
8. Prensacable PG11.

IMPORTANTE: para mantener el grado de protección IP55 del contenedor, utilizar racores tubo/caja, prensacables y (si corresponde) seccionadores rotativos que garanticen el mismo nivel de seguridad.

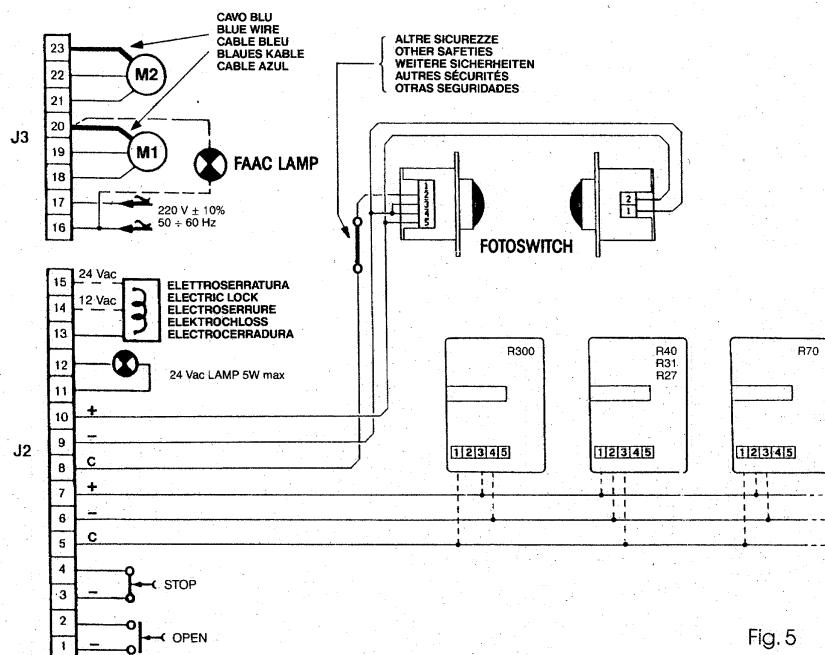


Fig. 5

FAAC 401 MPS

ELECTRONIC CONTROL UNIT

This electronic microprocessor unit is designed to control FAAC operators.

TECHNICAL CHARACTERISTICS

Power supply voltage	230 V~(+6 % -10 %) 50 (60) Hz
Absorbed power	35 W
Maximum load, motors	500 W
Max power, electric-lock	15 W a 12 V~
Max power, warning light	5 W - 24 Vcc
Max load, auxiliary equipment	500 mA
Temperature range	- 20° C + 55° C

INSTALLATION

The electronic control unit is housed inside a plastic box, IP 55 protection rating, designed to be fixed to the wall with screws or set into the wall.

ELECTRICAL CONNECTIONS

For safety reasons, a 16A differential switch with a threshold of 0,03A must be fitted upstream from the unit. When connecting up the unit, follow current safety and accident prevention regulations.

Separate cables must be used for power connections (power supply-motors) and for the command signal cables (push-buttons - receiver - photocells, etc.)

Warning: Switch of the power supply before carrying out any work on the equipment (connections, programming, maintenance, etc.)

TERMINAL BOARD CONNECTIONS (Fig. 5)

- 1) Common (—)
- 2) Open push-button (N.O.)
- 3) Common (—)
- 4) Stop push-button (N.C.)
- 5) Receiver contact (N.O.)
- 6) Common (—)
- 7) Power supply output + 30 V
- 8) Safety contact (N.C.)
- 9) Common (—)
- 10) Power supply output + 30 V
- 11) Warning light (24 Vac 5 W max)
- 12) Common (—)
- 13) Electric-lock, common
- 14) Electric-lock, 12 Vac
- 15) Electric-lock, 24 Vac
- 16) Power supply 220 V (neutral)
- 17) Power supply 220 V
- 18) Motor 1 forward
- 19) Motor 1 backward
- 20) Motore 1 common - blue cable
- 21) Motore 2 forward
- 22) Motor 2 backward
- 23) Motor 2 delayed common - blu cable

Note: if the stop push-button and/or the safety switches are not connected, their terminals must be bridged (3 with 4 and/or 8 with 9).

SURGE CAPACITOR CONNECTIONS (Fig. 1)

- J4) C1 capacitor for motor 1
J5) C2 capacitor for motor 2

COMPONENTS (Fig. 1)

- DL1 - Open push-button LED
DL2 - Safety LED
DL3 - Stop push-button LED
DS1 - Dip - switch group
F1 - 1.6 A anti-surge fuse (auxiliary equipment power supply)
F2 - 250 mA anti-surge fuse (H.V. supply)
F3 - 5A quick blow fuse (Motors)
J1 - Connector for DECODER card
J2 - Low voltage terminal board

- J3 - High voltage terminal board
LK1 - Electric-lock operating mode wire link.
LK2 - Warning light operating mode wire-link.

PROGRAMMING

The FAAC 401 MPS electronic control unit can be programmed using the microswitch unit and the three wire links (LK) to suit customer and system requirements.

PROGRAMMABLE FUNCTIONS

- 1) Operating logic
- 2) Opening/closing time
- 3) Pause time
- 4) Leaf delay time
- 5) Reversing stroke
- 6) Warning light operation mode
- 7) Electric-lock operation mode
- 8) 1 motor / 2 motor

1) Operating logics

The following logics are available:

- Logic A : "Automatic"
Logic E : "Semi-automatic"
Logic S1: "Safety"
Logic S2: "Safety Plus"

Use microswitches S8 and S9 to select the desired logic (Fig. 3)

Logic A: "AUTOMATIC"

When a command signal is transmitted (with push-button, transmitter, etc.) the motors open the gate and then stop at the end of the run time.

The gate stays open for the pause time, then closes automatically.

Any signal sent during the opening phase is not accepted. If a signal is sent during the pause time, the gate closes after 5 seconds.

If a signal is sent during the closing phase, the movement is reversed.

Logic E: "SEMI-AUTOMATIC"

The first signal opens the gate, which stays open. A second signal is needed to activate the closing phase.

A signal sent during the opening phase stops the movement of the gate, another signal actuates the closing phase.

If a signal is sent during the closing phase, the movement is reversed.

Logic S1: "SAFETY"

This logic functions in a similar way to logic A, the only difference being that a signal transmitted during the opening phase or during the pause time causes immediate closure.

Logic S2: "SAFETY PLUS"

This logic functions in the same way as logic S1, the only difference is the way the safety devices operate (see section "SAFETY DEVICES").

2) Opening/closing times

Microswitches S1-S2-S3-S4 are used to select one of the 16 different times available.

It is advisable to set a time which is slightly longer than the time taken by the system to reach the opening or closing position.

3) Pause time

Microswitches S5 and S6 are used to select one of the four available pause times, from a minimum of 0 secs to a maximum of 60 secs. (Fig. 3).

4) Leaf delay time

For double leaf swing gates, a leaf delay function is available to delay the closing of one leaf.

Microswitches S10 and S11 are used to select the desired leaf delay time, from a minimum of 0 secs to a maximum of 12 seconds.

5) Reversing stroke

This function, which can be activated using microswitch S7, is designed to facilitate the release of any electrolock connected up. When the signal is transmitted, a short closing phase is actuated, releasing the electric-lock and starting the opening phase.

6) Warning light operation mode

To display the state of the gate at a distance, there is a warning light output with a voltage of 24 Vac, max power 5W. Two operating modes can be selected using the wire link LK2 (Fig. 4).

Mode A (LK2 open): Light off - gate closed; light on continuously - gate is opening or in pause time; flashing light - gate closing.

Mode B (LK2 closed): Light off - gate closed; light on continuously - gate in opening, pause or closing phase.

7) Electric-lock operation mode

The FAAC 401 MPS unit is designed for the connection of electrolocks with 12 V and 24 V outputs (The 24 V output allows connection of two electric-locks in series for locking the gate in both open and closed positions.

Bridge LK1 (Fig. 2) is used to control electric-locks in the following ways.

Mode A (LK1 open): release when opening and closing.

Mode B (LK1 closed): release when opening only.

8) Number of motors

The unit is designed for the connection of 1 motor or 2 motors. The selection is achieved by means of the dip-switch S12.

IMPORTANT: Whenever gate/door movement is reversed, the motor power supply time is automatically reprogrammed to become the completed movement time plus 2 seconds.

This function is also actuated after a stop.

This function reduces by-pass run time and thus optimises operator functioning.

COMMAND FUNCTIONS

Open command

This includes any type of device (push-button, photocell, pneumatic pad, magnetic loop etc.) which transmits a signal to close a contact. The contact must be of the "normally open" type.

Stop command

Pressing the stop push-button causes an immediate stop in any situation.

By sending a new open signal, the closing phase is activated. The stop push-button contact must be of the normally closed type.

SAFETY DEVICES

These include all devices (photocells, safety edges, magnetic loops, etc.) designed to make the automatic system safer. The control panel has terminals for connection to these devices, which should have contacts of the normally closed type.

There is also a 30 Vcc output for supplying the photocells and any other safety device requiring a power supply.

Fig. 5 shows the connection of a pair of photocells with a

safety function. For further information on the use of photocells refer to the "PHOTOSWITCH" instructions.

Here is a description of how the safety devices operate in various conditions:

During the opening phase:

In this condition any triggering of safety devices is ignored.

During the pause time:

The pause time count is stopped while the safety device is in function, after which, the counting of the residual time starts again in logics A1 and S2, while in logic S1 the pause time is reset and the closing phase is activated after 5 seconds.

During the closing phase:

In logics A1, E1 and S1 the intervention of the photocells causes immediate reversal of movement.

In logic S2, the closing phase is stopped while the safety device remains in operation. The opening phase only begins when the obstacle is removed.

RADIO RECEIVER CONNECTIONS

Figure 5 shows the electrical connections with the various FAAC receiver models.

E RP 433 ESL / EDS DECODING CARD FACILITIES

The FAAC 401 MPS unit features quick connector facilities for the following cards:

- DECODER/MINIDEC - SL/DS in conjunction with PLUS 433 E multi-channel receiver;
- DECODER DS in conjunction with DIGICARD magnetic card drive and DIGIKEY keypad;
- RP 433 ESL/EDS receivers

The cards can be fitted in quick connector J1 (Fig. 1).

LED INPUT INDICATOR

To check input operation, the equipment is fitted with three LEDs which light up when the respective contact in the terminal board is closed.

This means that the LEDs for the inputs with N.C. contact, stop and photocell (DL3 and DL2 respectively) are normally lit, while LED DL1 (open push-button N.O.) lights up only when the contact is energised.

AUXILIARY EQUIPMENT POWER SUPPLY

Photocells, radio, receivers, card readers, keyboards and other auxiliary components can be connected to the equipment using the power supply output 30 Vcc.

For the system to function correctly and to prevent overloading, it is advisable to consult the following table to check that the total absorption of the auxiliary equipment connected up does not exceed the value of 300 mA.

TYPE OF ACCESSORY	NOMINAL CURRENT DRAWN
R31	50 mA
PLUS 433 E	25 mA
MINIDEC ESL / DS	4,5 mA
DECODER ESL / DS	30 mA
RP 433 ESL / EDS	36 mA
DIGICARD	15 mA
METAL DIGIKEY	15 mA
FOTOSWITCH	90 mA
DETECTOR F4 / PS6	50 mA
MINIBEAM	70 mA

MAINTENANCE

The equipment does not require maintenance.